

D547

Evidence-Based Healthcare Administration

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A hospital's quality committee, meeting in its newly renovated boardroom, wants the strongest evidence on whether hourly nurse rounding reduces patient falls. The committee chair has twenty years of bedside experience and strong views on rounding. Which source should carry the most weight?

- A. A systematic review and meta-analysis of randomized trials on hourly rounding.
- B. The chair's expert judgment, given two decades at the bedside.
- C. A cohort study of 50,000 patients, because a bigger sample always means stronger evidence.
- D. One large randomized trial run at a single academic medical center.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which source sits at the lowest level of the evidence hierarchy?

- A. A randomized controlled trial.
- B. An editorial giving a respected physician's opinion.
- C. A systematic review of randomized trials.
- D. A case-control study.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Data from 40 clinics show that clinics with more patient-portal logins have lower 30-day readmission rates. The data were pulled by a vendor's analytics team using a popular dashboard tool. Clinics with high portal use also serve younger, higher-income patients. What is the best conclusion?

- A. Lower readmissions cause patients to log in to the portal more.
- B. The finding is invalid because vendor dashboards are biased.
- C. The link may be driven by patient age and income; a controlled study is needed before claiming the portal cuts readmissions.
- D. Portal use lowers readmissions, so the system should push portal enrollment to cut readmissions.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A study of 120,000 discharge records found that a new scheduling tool cut average clinic wait time from 22.0 to 21.7 minutes, $p < .001$. The study was published in a well-regarded journal and funded by a state grant. The tool costs \$300,000 a year. What should the administrator conclude?

- A. The tool produces a large improvement, because $p < .001$.
- B. The result is not statistically significant, because 0.3 minutes is small.
- C. The difference is statistically significant but only 0.3 minutes, so its practical value must be weighed against the \$300,000 cost.
- D. The tool cut wait times by 22 minutes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A health system wants a model that predicts which discharged patients will be readmitted within 30 days. It has five years of records in which each patient is labeled readmitted or not readmitted. The data sit in a cloud warehouse the IT team migrated last spring. Which approach fits?

- A. Supervised learning, such as a classification model trained on the labeled outcomes.
- B. Unsupervised clustering, because the data set is large.
- C. No model is possible until the data move back on premises.
- D. A descriptive dashboard of past readmission rates by unit.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

An analyst runs an algorithm on patient records that have no outcome labels, and it sorts patients into five groups with similar service-use patterns. What kind of learning is this?

- A. Natural language processing.
- B. Reinforcement learning.
- C. Supervised classification.
- D. Unsupervised learning (clustering).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A clinic's new patient-satisfaction survey, printed on branded letterhead and offered in three languages, gives nearly identical scores when the same patients retake it a week later. Most of its questions ask about parking and waiting-room décor rather than the care patients received. Which statement fits?

- A. It is valid because it is offered in three languages.
- B. It is reliable but not a valid measure of satisfaction with care.
- C. It is valid, because it gives consistent results.
- D. It is valid but not reliable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A pain scale reflects patients' true pain well on average, but two nurses rating the same patient a few minutes apart often give very different scores. What is the main problem?

- A. Low validity.
- B. Sampling bias.
- C. The scale is useless and should never be used.
- D. Low reliability (poor inter-rater agreement).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A billing analyst at a 300-bed hospital that just earned Magnet status asks the data team for a report to check coding accuracy on cardiac claims. The data team could easily send the full medical record for each patient. Under HIPAA's minimum necessary standard, what should they send?

- A. Only the fields needed to check coding, such as encounter ID, diagnosis and procedure codes, and claim amounts.
- B. A fully de-identified file with all 18 identifiers removed, even though claims must be matched to encounters.
- C. The full medical records, because the analyst is an employee of the hospital.
- D. Nothing, because HIPAA bars all internal use of PHI without written patient consent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A sepsis alert built into the EHR by a well-known vendor was tested on 1,000 patients. Of the 100 patients who truly had sepsis, it flagged 80. Of the 900 without sepsis, it wrongly flagged 90. The pilot ran on three units over six weeks. What is the alert's sensitivity?

- A. 8%
- B. 80%
- C. 47%
- D. 90%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

11. [TRANSFER — reversed stem / different variable]

A screening tool has 95% specificity. Among 2,000 people who do not have the condition, how many will it wrongly flag as positive?

- A. 0, because 95% specificity is very high.
- B. 1,900
- C. 95
- D. 100

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12.

An administrator wants to know whether average length of stay, measured in days, differs between two hospitals in the same system: one uses a new discharge checklist and the other does not. Both hospitals were built in the 1990s. Different patients are treated at each site. Which test fits best?

- A. A chi-square test.
- B. A paired t-test.
- C. An independent-samples t-test.
- D. Descriptive statistics only, because inferential tests always require a randomized trial.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A hospital's quality committee, ~~meeting in its newly renovated boardroom,~~ wants the strongest evidence on whether hourly nurse rounding reduces patient falls. ~~The committee chair has twenty years of bedside experience and strong views on rounding.~~ Which source should carry the most weight?

Target word: strongest level of evidence **Decides:** Systematic reviews of randomized trials top the evidence hierarchy.

A	Correct	Pooling many randomized trials gives the least biased, most precise estimate.
B	Camouflage	Expert opinion is the lowest level of evidence, however experienced the expert.
C	Absolute language	Sample size does not fix the confounding built into an observational design.
D	Adjacent category	A single trial is strong but sits one level below a systematic review of many trials.

2. Answer B Concept 1 • transfer

Which source sits at the lowest level of the evidence hierarchy?

Target word: bottom of the hierarchy — reverse direction **Decides:** Expert opinion without systematic data ranks last.

A	Confusion set	Trials rank near the top because randomization controls confounding.
B	Correct	Opinion not based on systematically gathered data is the weakest evidence.
C	Inverted operation	That is the top of the hierarchy, not the bottom.
D	Adjacent category	Observational and weaker than trials, but still above opinion.

3. Answer C Concept 2

Data from 40 clinics show that clinics with more patient-portal logins have lower 30-day readmission rates. The data were pulled ~~by a vendor's analytics team using a popular dashboard tool.~~ Clinics with high portal use also serve younger, higher-income patients. What is the best conclusion?

Target word: what the association supports **Decides:** Portal use travels with age and income, so confounding is possible.

A	Inverted operation	Reverses the direction without evidence; the real issue is confounding.
B	Camouflage	Who ran the query does not decide the issue; the study design does.
C	Correct	Observational data with obvious confounders cannot establish cause.
D	Confusion set	Treats correlation as causation.

4. Answer C Concept 3

A study of 120,000 discharge records found that a new scheduling tool cut average clinic wait time from 22.0 to 21.7 minutes, $p < .001$. The study was ~~published in a well-regarded journal and funded by a state grant~~. The tool costs \$300,000 a year. What should the administrator conclude?

Target word: significance vs. size of effect **Decides:** The effect is 0.3 minutes, tiny despite $p < .001$.

A	Confusion set	A p-value measures how unlikely chance is, not how big the effect is.
B	Adjacent category	Mixes practical size with statistical significance; $p < .001$ is significant.
C	Correct	Large samples make tiny effects significant; practical importance is a separate judgment.
D	Wrong stem number	22.0 is the starting wait, not the change.

5. Answer A Concept 4

A health system wants a model that predicts which discharged patients will be readmitted within 30 days. It has five years of records in which each patient is labeled readmitted or not readmitted. ~~The data sit in a cloud warehouse the IT team migrated last spring~~. Which approach fits?

Target word: type of machine learning **Decides:** The outcome is labeled and the goal is to predict it.

A	Correct	Known labels plus a prediction goal define supervised classification.
B	Confusion set	Size does not decide the method; labeled outcomes do.
C	Camouflage	Where the data are stored does not decide the learning type.
D	Adjacent category	Describes the past but does not predict individual patients.

6. Answer D Concept 4 · transfer

An analyst runs an algorithm on patient records that have no outcome labels, and it sorts patients into five groups with similar service-use patterns. What kind of learning is this?

Target word: name the method from the result **Decides:** No labels; groups found from the data.

A	Adjacent category	NLP works on text; nothing here involves language.
B	Confusion set	Reinforcement learning learns from rewards over repeated actions.
C	Inverted operation	Classification needs known labels to learn from.
D	Correct	Grouping unlabeled records by similarity is clustering.

7. Answer B Concept 5

A clinic's new patient-satisfaction survey, ~~printed on branded letterhead and offered in three languages~~, gives nearly identical scores when the same patients retake it a week later. Most of its questions ask about parking and waiting-room décor rather than the care patients received. Which statement fits?

Target word: reliability vs. validity **Decides:** Consistent on retest, but the items do not measure care satisfaction.

A	Camouflage	Language access matters, but it does not make the questions measure care.
B	Correct	Stable scores show reliability; off-target questions undermine validity.
C	Confusion set	Consistency is reliability, not validity.
D	Inverted operation	Swaps the two; the retest result shows it is reliable.

8. Answer D Concept 5 · transfer

A pain scale reflects patients' true pain well on average, but two nurses rating the same patient a few minutes apart often give very different scores. What is the main problem?

Target word: which property fails — reverse case **Decides:** Scores vary between raters for the same patient.

A	Inverted operation	The stem says it tracks true pain on average; consistency is what fails.
B	Adjacent category	Sampling bias concerns who is studied, not how consistently a tool scores.
C	Absolute language	Rater training can improve reliability; the tool is not worthless.
D	Correct	Inconsistent scores for the same patient are a reliability failure.

9. Answer A Concept 6

A billing analyst at a 300-bed hospital ~~that just earned Magnet status~~ asks the data team for a report to check coding accuracy on cardiac claims. The data team could easily send the full medical record for each patient. Under HIPAA's minimum necessary standard, what should they send?

Target word: scope of the disclosure **Decides:** The task is coding review; only coding-related fields are needed.

A	Correct	Minimum necessary limits PHI to what the task requires.
B	Adjacent category	De-identification is a different tool and would block the task.
C	Dropped element	Employment does not remove the minimum necessary limit on operations uses.
D	Absolute language	Payment and operations uses are permitted without authorization.

10. Answer B Concept 7

A sepsis alert built into the EHR ~~by a well-known vendor~~ was tested on 1,000 patients. Of the 100 patients who truly had sepsis, it flagged 80. Of the 900 without sepsis, it wrongly flagged 90. ~~The pilot ran on three units over six weeks.~~ What is the alert's sensitivity?

Target word: sensitivity **Decides:** True positives / all who have sepsis = 80 / 100.

A	Wrong stem number	Divides by all 1,000 patients instead of the 100 with sepsis.
B	Correct	80 of the 100 true sepsis cases were flagged.
C	Adjacent category	That is positive predictive value: 80 / 170 flagged.
D	Confusion set	That is specificity: 810 true negatives / 900.

11. Answer D Concept 7 · transfer

A screening tool has 95% specificity. Among 2,000 people who do not have the condition, how many will it wrongly flag as positive?

Target word: false positives from specificity **Decides:** False-positive rate = 1 – 0.95 = 5%.

A	Absolute language	Any specificity below 100% produces some false positives.
B	Inverted operation	That is the true negatives the tool correctly clears.
C	Wrong stem number	Uses the percentage as a count.
D	Correct	5% of 2,000 = 100 false positives.

12. Answer C Concept 8

An administrator wants to know whether average length of stay, measured in days, differs between two hospitals in the same system: one uses a new discharge checklist and the other does not. ~~Both hospitals were built in the 1990s.~~ Different patients are treated at each site. Which test fits best?

Target word: choose the statistical test **Decides:** Continuous outcome, two independent groups.

A	Confusion set	Chi-square compares counts in categories, not means of days.
B	Adjacent category	Paired tests need the same subjects measured twice.
C	Correct	Compares the means of a continuous measure between two separate groups.
D	Absolute language	Inferential tests apply to observational comparisons too, with care about confounding.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	1	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The evidence hierarchy	Systematic reviews and meta-analyses of randomized trials sit at the top; single trials, cohort and case-control studies follow; expert opinion sits at the bottom.	Giving the most weight to the most senior person in the room or the biggest single study.	A list of sources and the words strongest or most weight.
2	Correlation is not causation	An association in observational data can come from confounders or reverse direction; causal claims need a controlled design.	Launching a program because two numbers move together.	Groups that differ in more than one way (age, income, location) plus a claim that one thing causes another.
3	Statistical vs. practical significance	A small p-value says a difference is unlikely to be chance; it says nothing about whether the difference is big enough to matter.	Reading $p < .001$ as a large or important effect.	A huge sample, a tiny difference, and a very small p-value.
4	Supervised vs. unsupervised machine learning	Supervised learning trains on labeled outcomes to predict them; unsupervised learning finds structure (such as clusters) in unlabeled data.	Calling any big-data model clustering, or any model with an output supervised.	Whether the records already carry a known outcome label.
5	Reliability vs. validity of instruments	Reliability is consistency (same result on repeat); validity is accuracy (measures what it claims to measure). A tool can be reliable without being valid.	Assuming a consistent instrument must be measuring the right thing.	Repeat scores match, but the questions are about something else, or the reverse.
6	HIPAA minimum necessary	Uses and disclosures of PHI for operations and payment must be limited to the minimum needed for the task.	Assuming internal staff can see the full record, or that HIPAA forbids all internal use.	A staff member with a specific job asks for patient data.
7	Sensitivity and specificity	Sensitivity = true positives / all who have the condition; specificity = true negatives / all who do not.	Dividing by the whole sample or swapping sensitivity with positive predictive value.	A 2×2 set of counts: truly sick vs. not, flagged vs. not.
8	Matching the test to the data	Compare means of a continuous measure between two independent groups with an independent-samples t-test; use chi-square for counts in categories and a paired test for the same subjects measured twice.	Using chi-square on continuous data or a paired test on two separate groups.	The measurement scale (days, dollars vs. yes/no) and whether the groups are the same people.

Course scope (public description): Evidence-Based Healthcare Administration is an immersive course that equips students with the knowledge and skills needed to apply evidence-based practices in the field of healthcare administration. Students will gain a thorough understanding of how to use data analytics, research methodologies, and evidence-based decision-making principles. The course covers essential areas, such as machine learning, artificial intelligence, data collection instruments, and statistical and analytical tools. Ethical considerations, privacy regulations, and effective communication skills are also highlighted, preparing students to make informed

decisions, advocate for evidence-based strategies, and contribute to improved patient care, better outcomes, and sustainable healthcare practices.